INCH-POUND
MIL-DTL-15291/1B
6 February 2006
SUPERSEDING
MIL-S-15291/1A(SH)
22 July 1991

DETAIL SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 1SR FRONT MOUNTED, BACK CONNECTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-15291.

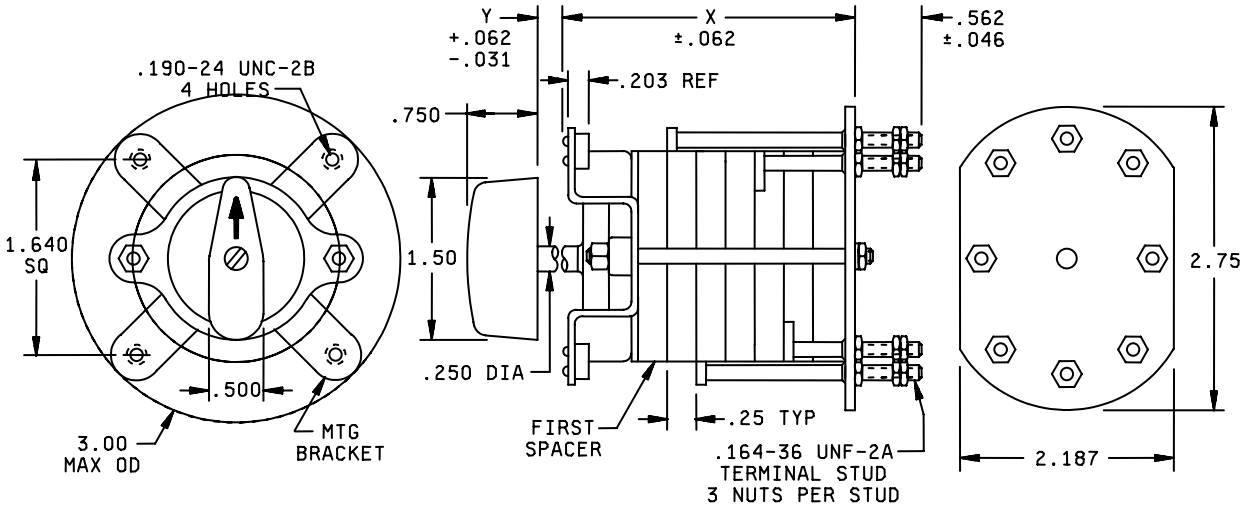



FIGURE 1. Class 1SR switch (up to 6 studs).

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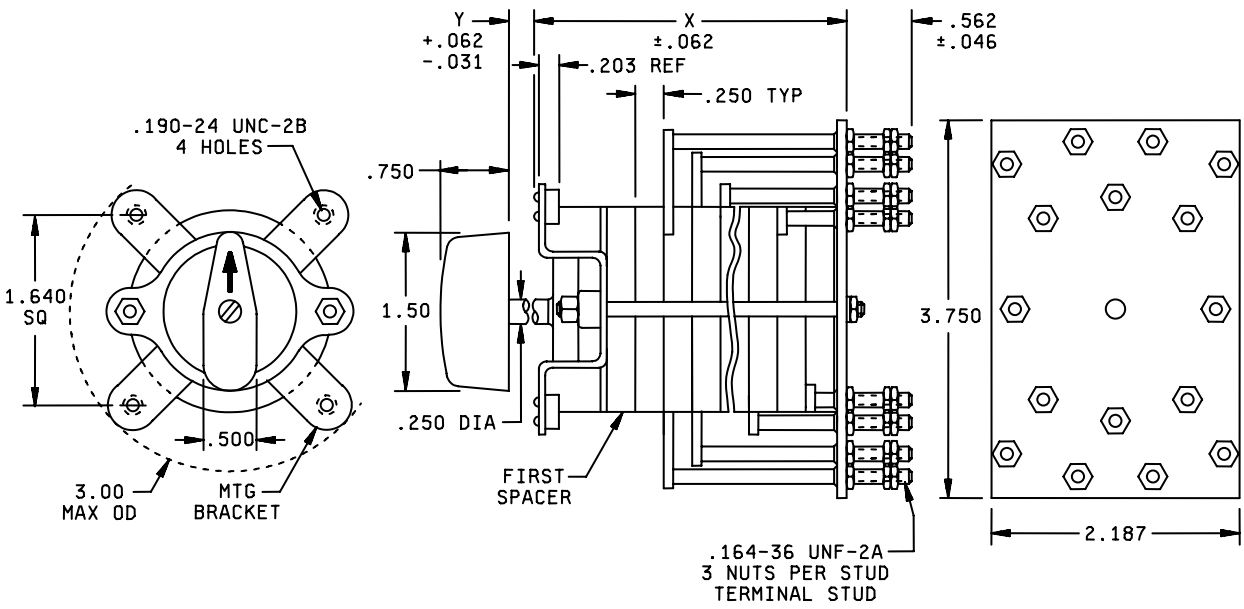



FIGURE 2. Class 1SR switch (up to 14 studs).



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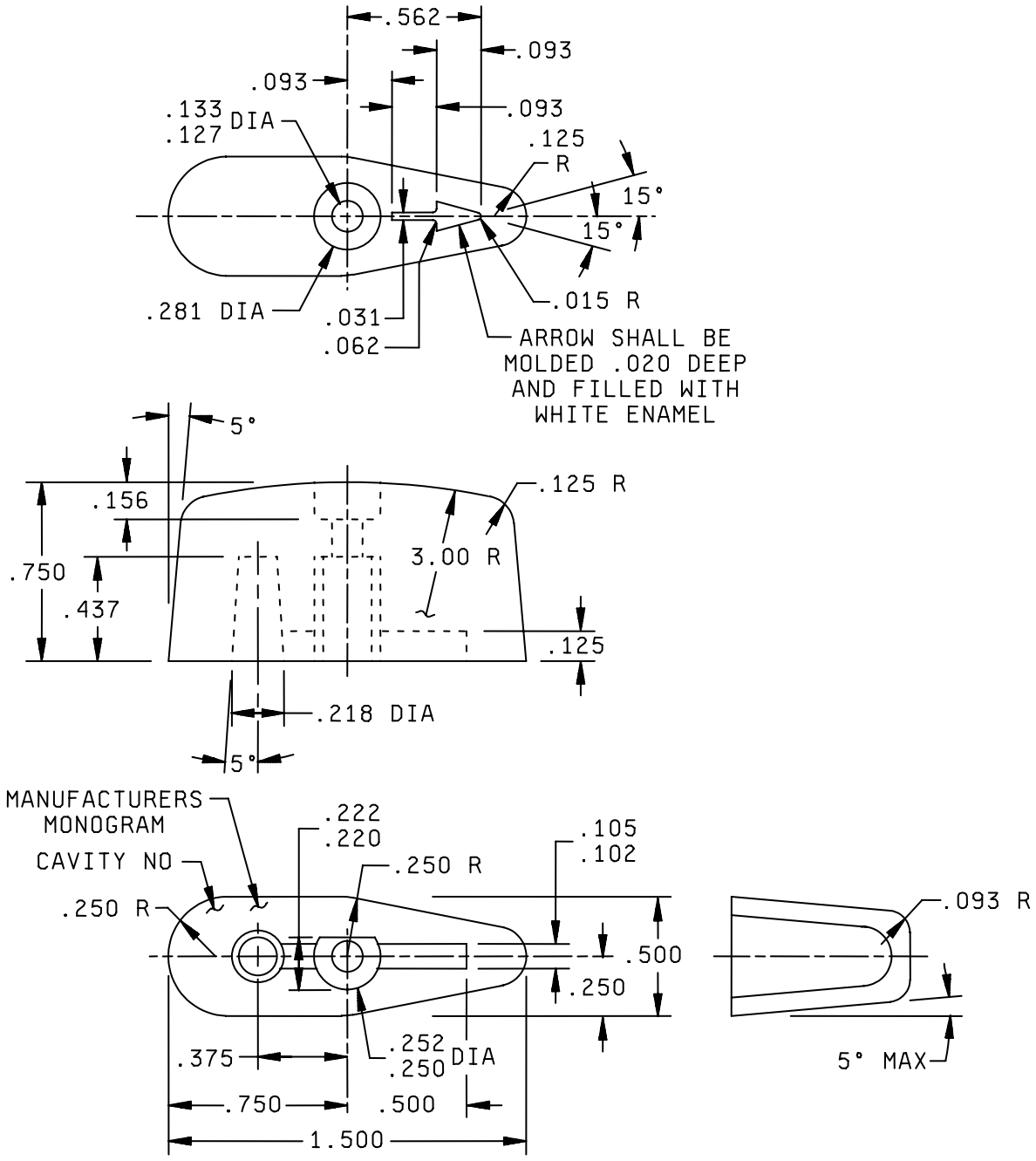



FIGURE 3. Handle.



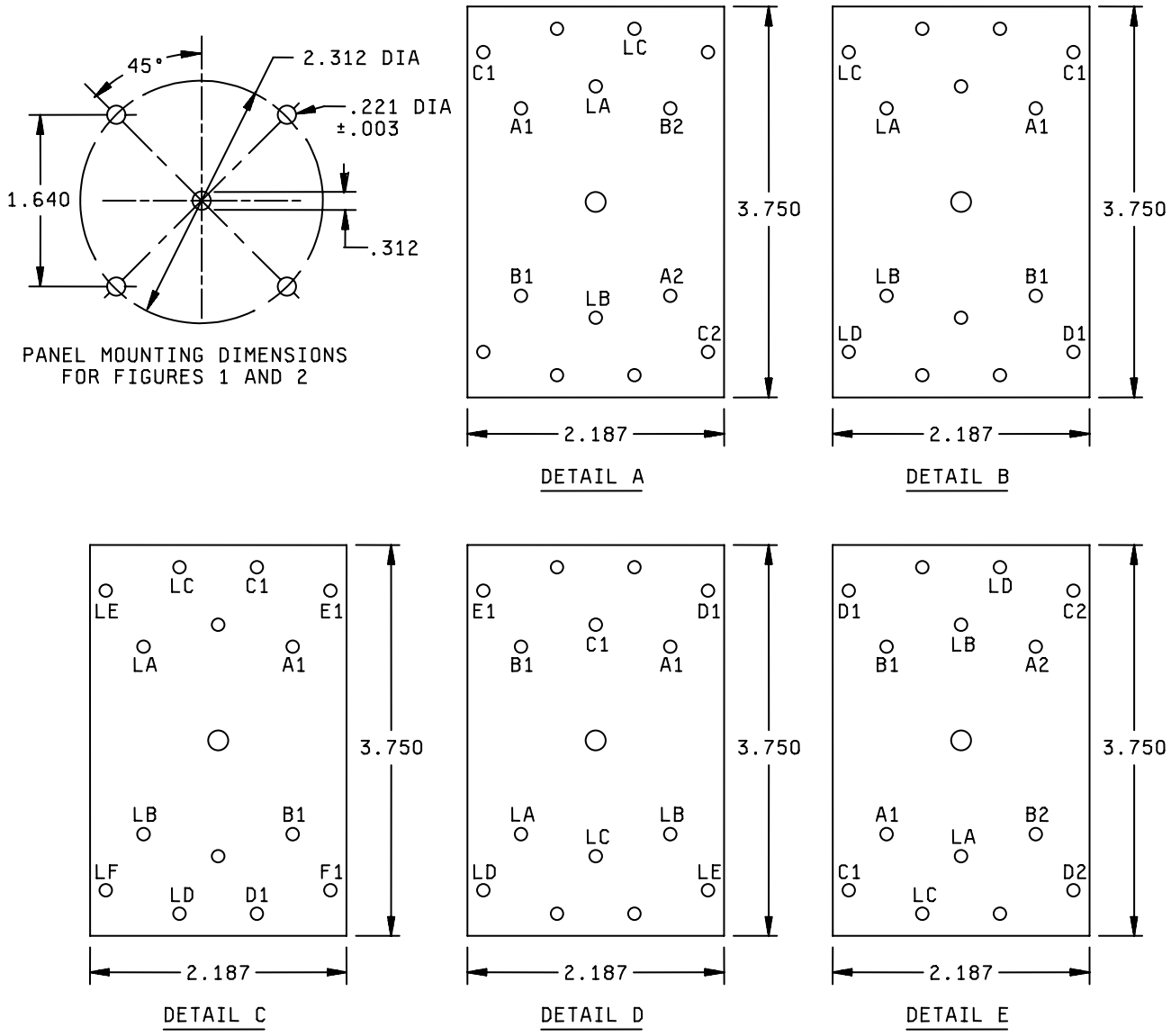
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TABLE I. Type and switching characteristics.

| M15291/1 DASH NO. TYPE DESIGNATION | DETAIL REF | DIM "X" | HANDLE OR SHAFT POSITION | CIRCUIT AND SPACER CONFIGURATIONS ROTOR POSITION, SPACER LOCATIONS, TERMINAL MARKING AND LOCATIONS | | | | | | | | | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | TORQUE IN-LBS | DIM "Y" | | NO 1 | NO 2 | NO 3 | NO 4 | NO 5 | NO 6 | NO 7 | NO 8 | NO 9 | |
| -001 | FIG 1 | 1.687 | OFF ON OFF | | B1 LB | A1 LA | | | | | | | |
| 1SR2A1 | 4 | .437 | | | | | | | | | | | |
| -002 | FIG 1 | 1.937 | OFF ON OFF | | C1 LC | B1 LB | A1 LA | | | | | | |
| 1SR3A1 | 4 | .437 | | | | | | | | | | | |
| -003 | FIG 2 DETAIL D | 2.187 | OFF ON OFF | | D1 LD | C1 LC | B1 LB | A1 LA | | | | | |
| 1SR4A1 | 4 | .437 | | | | | | | | | | | |
| -004 | FIG 2 DETAIL D | 2.437 | OFF ON OFF | | E1 LE | D1 LD | C1 LC | B1 LB | A1 LA | | | | |
| 1SR5A1 | 4 | .437 | | | | | | | | | | | |
| -005 | FIG 2 DETAIL A | 2.687 | OFF ON 1 OFF | | LC | C1 C2 | B2 B1 | LB | LA | A1 A2 | | | |
| 1SR3B1 | 4 | .437 | | | | | | | | | | | |
| -006 | FIG 2 DETAIL E | 3.187 | OFF ON 1 OFF | | LD | D1 D2 | C2 C1 | LC | LB | B1 B2 | A2 A1 | LA | |
| 1SR4B1 | 4 | .437 | | | | | | | | | | | |
| -007 | FIG 2 DETAIL B | 2.187 | OFF ON 1 OFF | | D1 LD | C1 LC | B1 LB | A1 LA | | | | | |
| 1SR4F1 | 4 | .437 | | | | | | | | | | | |
| -008 | FIG 2 DETAIL C | 2.687 | OFF ON 1 ON 1&2 | | F1 LF | E1 LE | D1 LD | C1 LC | B1 LB | A1 LA | | | |
| 1SR6F1 | 4 | .437 | | | | | | | | | | | |



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NOTE: Viewed from the rear of the switch.

FIGURE 4. Mounting dimensions and terminal stud locations.



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REQUIREMENTS:

Applicable specification: MIL-DTL-15291.

Dimensions and mounting: See figures 1 through 4.

Electrical circuits: See table I.

Angle of throw: 90 degrees.

Switching action: Snap action, reciprocating.

Electrical and endurance ratings: See table II.

Stop strength (applicable to switches with stops): 30 inch-pounds.

Vibration: 50 Hz, MIL-STD-167-1

Shock: High impact, MIL-S-901

Overload: 600 percent of ac rating, 50 make and break operations.

Contact resistance: 0.02 ohms maximum.

Dielectric withstanding voltage: 1,250 V rms.

Insulation resistance: 200 megohms minimum.

Temperature rise: 50°C maximum.

Mounting screws: .190-24UNC-2A (4) length and headstyle to suit application. Screws not furnished.

TABLE II. Electrical and endurance ratings.

| Tests | Current (amperes) | Voltage (volts) | Electrical operations (number of operations) | Test rate (operations per minute) | De-energized operations (number of operations) |
|---|---|---|---|---|---|
| Alternating current (rms) | 10 | 125 | 30,000 | 30 | 20,000 |
| Direct current | 10 | 120 | 20,000 | 15 | --- |

Terminal marking: Terminal markings shown in table I locate terminals as viewed from the front of the switches.

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended rating listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.



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TABLE III. Extended ratings.

| Switching characteristics | AC - 60 or 400 Hertz | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 125 volts | | | | 250 volts | | | | 500 volts | | | |
| | Resistive or lamp load | | Inductive load 0.75 pF | | Resistive or lamp load | | Inductive load 0.75 pF | | Resistive or lamp load | | Inductive load 0.75 pF | |
| | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations |
| All | 10 | 30,000 | 10 | 30,000 | --- | --- | --- | --- | --- | --- | --- | --- |
| F 1/ | 15 | 10,000 | 15 | 10,000 | 10 | 10.000 | 10 | 10,000 | --- | --- | --- | --- |
| A, B 2/ | 15 | 10,000 | 15 | 10,000 | 10 | 10,000 | 10 | 10,000 | 5 | 10,000 | 5 | 10,000 |
| A, B | 15 | 10,000 | 15 | 10,000 | 10 | 10,000 | 15 | 10,000 | 10 | 10,000 | 10 | 10,000 |
| Switching characteristics | DC | | | | | | | | | | | |
| | 120 volts | | | | 250 volts | | | | 350 volts | | | |
| | Resistive or lamp load | | Inductive, load 3/ | | Resistive or lamp load | | Inductive, load 3/ | | Resistive or lamp load | | Inductive, load 3/ | |
| | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations | Amp | Operations |
| F | 10 | 10,000 | 10 | 10,000 | 5 | 10,000 | 5 | 10,000 | 2 | 10,000 | 2 | 10,000 |
| F 2/ | --- | --- | --- | --- | --- | --- | --- | --- | 1 | 10,000 | 1 | 10,000 |
| A, B 2/ | 10 | 10,000 | 10 | 10,000 | 5 | 10,000 | 5 | 10,000 | 2 | 10,000 | 2 | 10,000 |
| A, B | 10 | 20,000 | 15 | 10,000 | --- | --- | --- | --- | --- | --- | --- | --- |

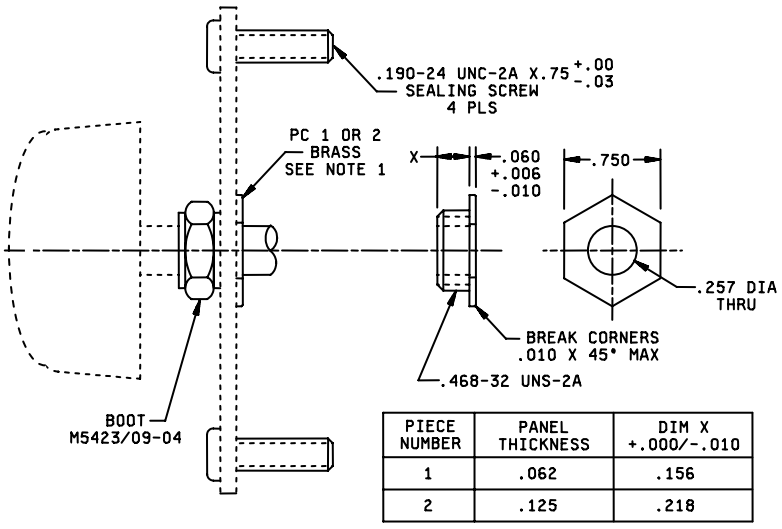
1/ Switching characteristic F is not rated for 500 volts in class 1SR.
2/ Single pole break - all other ratings are based on breaking both sides of the line in accordance with figures 1 and 2 of MIL-DTL-15291.
3/ 0.24 henry for inductive current.





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Dripproof mounting: Switches listed in table I as having four-polar mounting and "Y" dimensions not less than 7/16 inch can be mounted with a kit (see figure 5), to affect a dripproof seal between switch and mounting panel. Panel drilling must comply with figure 6. Kits not supplied unless specified (see 6.2 of MIL-DTL-15291.)

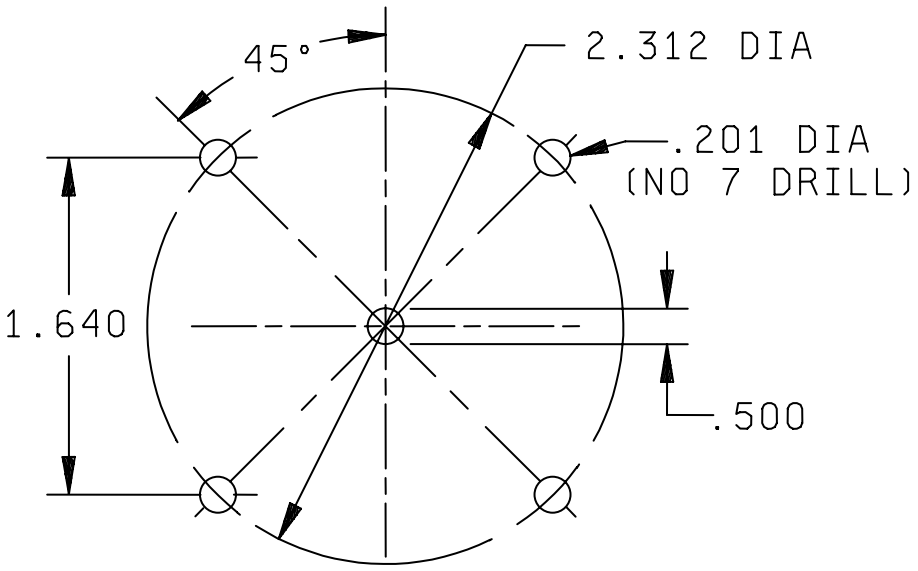



| PIECE NUMBER | PANEL THICKNESS | DIM X +.000/-.010 |
|---|---|---|
| 1 | .062 | .156 |
| 2 | .125 | .218 |

FIGURE 5. Kit for panel seal mounting.



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NOTE: Do not chamfer mounting holes. Breaking 1/64 maximum permissible.

FIGURE 6. Panel drilling for mounting kit.



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Application and Acquisition Guide: PIN and type designation cross reference shall be as shown on table IV.

TABLE IV. Application and acquisition guide.

| M15291/1 dash no. | Type designation | For new or existing design | For replacement | Circuit configuration |
|---|---|---|---|---|
| -001 | 1SR2A1 | X | | A (off-on-off-on) |
| -002 | 1SR3A1 | X | | A (off-on-off-on) |
| -003 | 1SR4A1 | X | | A (off-on-off-on) |
| -004 | 1SR5A1 | --- | | A (off-on-off-on) |
| -005 | 1SR3B1 | --- | | B (off-on-off-on 2) |
| -006 | 1SR4B1 | --- | | B (off-on 1-off-on 2) |
| -007 | 1SR4F1 | --- | | F (off-on 1-on 1 and 2-on 2) |
| -008 | 1SR6F1 | --- | | F (off-on 1-on 1 and 2-on 2) |

Referenced Documents:
MIL-DTL-15291
MIL-S-901
MIL-STD-167-1

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-1897)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .



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